

f.i. $+\infty - \infty$, $0 \cdot \infty$, $\frac{0}{0}$, $\frac{\infty}{\infty}$

~~$$\frac{2}{3} \cdot \frac{2}{+\infty} = 0$$~~ ERRORE

Si scrive così: $\frac{2}{m} \xrightarrow{D \rightarrow \infty} 0$

Prop $f: X \rightarrow \mathbb{R}$ $x \in \mathbb{R}$ f funkt elementore
 $\{a_n\} \subseteq X$ $a_n \rightarrow l \in X$

es: $\frac{n}{2n+1} \rightarrow \frac{1}{2}$ (vedremo il perché)

Limiti di sup e inf mediante le funzioni elementari

Superintense m^d $d \in \mathbb{R}$

$$d = 0 \quad z^0 = 1 - \Delta 1$$

$d = 0$ $3^d = 1 \rightarrow 1$
 $d > 0$ $3^d \rightarrow +\infty$ infatti $n^d > K \Leftrightarrow n > K^{\frac{1}{d}}$ D. vero

$$d < 0 \quad m^2 \rightarrow 0 \quad \Rightarrow \quad m^2 = \frac{1}{m^{-2}}$$

Polinomio

$$x_m = q_0 m^r + q_1 m^{r-1} + \dots + q_{r-1} m + q_r \quad \begin{matrix} q_1, q_2, \dots, q_r \in \mathbb{R} \\ r \in \mathbb{N} \end{matrix}$$

e secondo dei regimi dei coeff si può avere una f.i. $+\infty - \infty$

$$(es: \underset{+\infty}{3} \underset{+\infty}{m^4} - \underset{+\infty}{6} \underset{+\infty}{m^3} + \underset{+\infty}{2} \underset{+\infty}{m^2} - \underset{+\infty}{5} \underset{+\infty}{m} + \underset{+\infty}{8})$$

$$X_m = m^r \left(\frac{e_m}{m} + \frac{e_1}{m^2} + \frac{e_2}{m^3} + \dots + \frac{e_r}{m^r} \right)$$

$$\text{es: } \underset{+\infty}{\underset{\downarrow}{n^4}} \left(\underset{+\infty}{\underset{\downarrow}{3}} - \underset{0}{\underset{\downarrow}{\frac{6}{n}}} + \underset{0}{\underset{\downarrow}{\frac{2}{n^2}}} - \underset{0}{\underset{\downarrow}{\frac{5}{n^3}}} + \underset{0}{\underset{\downarrow}{\frac{3}{n^4}}} \right) \rightarrow +\infty$$

$$x_n \rightarrow \begin{cases} +\infty & \text{se } a_0 > 0 \\ -\infty & \text{se } a_0 < 0 \end{cases}$$

I polinomi di grado sempre. Il segno della differenza è il segno del coefficiente del termine di grado massimo

$$\text{es: } n^3 - 2n^2 - 6n - 3 \rightarrow +\infty$$

$$n^3 + 2n^2 + 6n - 3n^4 \rightarrow -\infty$$

Funzioni razionali fratte

$$x_n = \frac{a_0 n^p + \dots + a_p}{b_0 n^q + \dots + b_q}$$

$$a_i, b_j \in \mathbb{R} \quad b_0 \neq 0 \\ p \in \mathbb{N}_0 \quad q \in \mathbb{N}$$

è una f. i. $\frac{\infty}{\infty}$

$$x_n = \frac{n^p \left(a_0 + \dots + \frac{a_p}{n^p} \right)}{n^q \left(b_0 + \dots + \frac{b_q}{n^q} \right)} = \underset{\downarrow}{n^{p-q}} \frac{a_0 + \dots + \frac{a_p}{n^p}}{b_0 + \dots + \frac{b_q}{n^q}}$$

$$\begin{cases} 1 & \text{se } p=q \\ +\infty & \text{se } p>q \\ 0 & \text{se } p<q \end{cases}$$

$$\text{quindi: se } p=q \quad x_n \rightarrow \frac{a_0}{b_0} \\ \text{se } p>q \quad x_n \rightarrow \infty$$

$+\infty$ se a_0 e b_0 sono concordi, $-\infty$ se discordi)

$$\text{se } p<q \quad x_n \rightarrow 0$$

$$\text{es: } \frac{3m^2 - 5m + 1}{(m+2)^2} \rightarrow 3$$

$$\frac{-2m^2 - 6m}{3m^4 + 5m^2 + 4} \rightarrow -\frac{2}{3}$$

$$\frac{3m^2 - 5m + 1}{(m+2)^4} \rightarrow 0$$

$$\frac{3m - 5m + 1}{m - 3} \rightarrow \infty$$

$$\frac{3m^2 - 5m^4 + 1}{m^2 + 8} \rightarrow -\infty$$

$$\frac{3m^2 - 5m^4 + 1}{m^2 - 8m^3 + 3} \rightarrow \infty$$

Successione geometrica

$$q \in \mathbb{R} \quad \{q^n\}$$

$$q = 0 \quad q^n \rightarrow 0$$

$$q = 1 \quad q^n \rightarrow 1$$

$$q = -1 \quad q^n \text{ oscilla.}$$

$$q > 1 \quad q^n \rightarrow +\infty \text{ m.fatti } q^n > k \Leftrightarrow n > \log_q k$$

$$q < -1 \quad q^n = ((-1)(-e))^n = \underbrace{(-1)^n}_{\text{oscilla } +\infty} \underbrace{(-e)^n}_{(-e) > 1} \text{ oscilla}$$

$$-1 < q < 1 \quad q^n \rightarrow 0 \text{ m.fatti } |q^n| < \varepsilon \Leftrightarrow n > \log_{|q|} \varepsilon \text{ vero } \triangleright \text{perché } |q| < 1$$

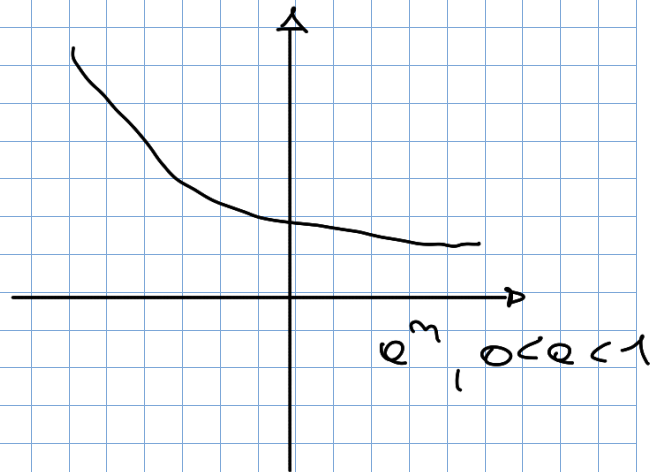
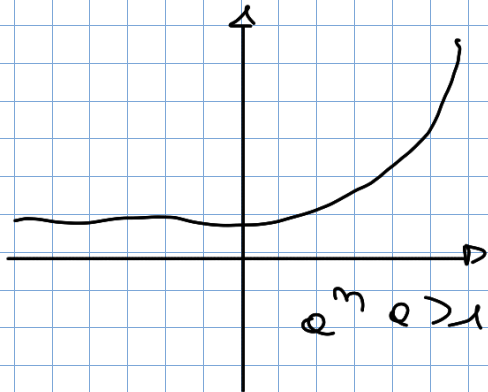
$$\text{es: } \left(-\frac{1}{5}\right)^n \rightarrow 0 \quad (-6)^n \text{ oscilla} \quad (18)^n \rightarrow +\infty$$

$$\left(\frac{2}{3}\right)^n \rightarrow 0$$

Successione composte mediante la funzione esponenziale

$\{x_n\}$ regolare con e^{x_n} con $e > 0, e \neq 1$

$$\text{se } x_n \rightarrow l \in \mathbb{R} \rightarrow e^{x_n} \rightarrow e^l$$



se $e > 1$

$$e^{x_n} \begin{cases} \rightarrow +\infty & \text{se } x_n \rightarrow +\infty \\ \rightarrow 0 & \text{se } x_n \rightarrow -\infty \end{cases}$$

se $0 < e < 1$

$$e^{x_n} \begin{cases} \rightarrow 0 & \text{se } x_n \rightarrow +\infty \\ \rightarrow +\infty & \text{se } x_n \rightarrow -\infty \end{cases}$$

es:

$$2 \quad \left(\frac{3n^2 - 1}{n + 4} \right) \begin{cases} \rightarrow +\infty \\ \rightarrow +\infty \end{cases}$$

$$\left(\frac{1}{3} \right) \left(\frac{n^6 - 2}{3 - n} \right) \begin{cases} \rightarrow -\infty \\ \rightarrow +\infty \end{cases}$$

$$3 \quad \left(\frac{n^6 - 6}{2 - n^5} \right) \begin{cases} \rightarrow -\infty \\ \rightarrow 0 \end{cases}$$

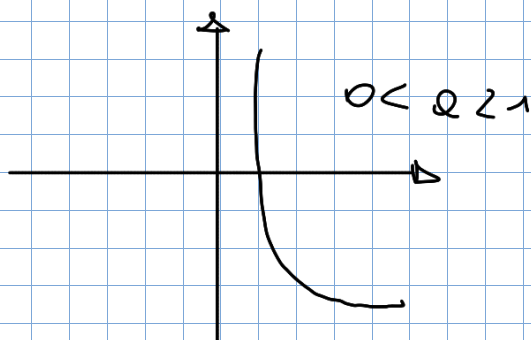
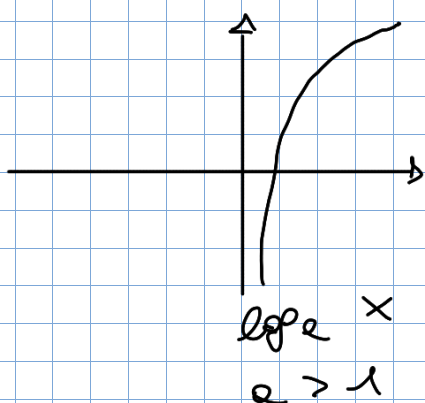
Funzione logaritmica

$\{b_m\}$ succ. regolare $b_m > 0 \quad \forall m \quad e > 0, e \neq 1$

$$x_m = \log_e b_m$$

$$\text{se } b_m \rightarrow l > 0 \Rightarrow x_m \rightarrow \log_e l$$

$$\text{es: } \log_2 \frac{4m+1}{m+3} \rightarrow \log_2 4 = 2$$



$$\text{se } e > 1 \Rightarrow x_m \begin{cases} +\infty & \text{se } b_m \rightarrow +\infty \\ -\infty & \text{se } b_m \rightarrow 0 \end{cases}$$

$$\text{se } 0 < e < 1 \Rightarrow x_m \begin{cases} -\infty & \text{se } b_m \rightarrow +\infty \\ +\infty & \text{se } b_m \rightarrow 0 \end{cases}$$

$$\text{es: } \log_3 \frac{2m+1}{m^2+8} \rightarrow -\infty \quad \log_2 \frac{2m^6+1}{m^4+8} \rightarrow +\infty$$

$$\log_{\frac{1}{2}} \left(\frac{1-3m^4}{2-m^2} \right) \xrightarrow{+\infty} -\infty$$

$$\log_{\frac{1}{4}} \left(\frac{3m+1}{2m^4+6} \right) \xrightarrow{0} +\infty$$

per es se $n \rightarrow +\infty, a > 1 \Rightarrow e^{x^n} \rightarrow +\infty$

si deve essere $e^{x^n} > k \Leftrightarrow x^n > \log_m k$ vero $\mathbb{D}$

se $0 < a < 1$ $e^{x^n} = \frac{1}{(\frac{1}{a})^{x^n}} \rightarrow 0$
 $\frac{1}{a} > 1$

se $b_n \rightarrow +\infty, a > 1 \Rightarrow \log_e b_n \rightarrow +\infty$

si deve essere $\log_e b_n > k \Leftrightarrow b_n > e^k$ vero $\mathbb{D}$

se $0 < a < 1$ $\log_e b = (\log_e \frac{1}{a})(\log_{\frac{1}{a}} b_n) \rightarrow -\infty$

se $a_n \rightarrow \infty$ allora $\cos a_n, \sin a_n, \tan a_n$ sono non regolari

Consideriamo ancora una successione del tipo con $a_n b^n$
con $a_n > 0$ $a_n \neq 1 \forall n$ $\{a_n\}, \{b_n\}$ regolare

Ricordiamo che se $x > 0$ $x = e^{\log x}$ quindi:

$$a_n b_n = e^{\log a_n b_n} = e^{b_n \log a_n}$$

$$\Rightarrow: a_n \rightarrow 1, b_n \rightarrow 3 \quad e^{3 \log a_n} \rightarrow e^{3 \log 2} = e^{\log 8} = 8$$

Si vuole un f.i. se cioè una f.i. nel prodotto
 $b_n \log a_n$

$$b_n \rightarrow$$

f.i. $0^0 (+\infty)^0$

$\log a_n \rightarrow \infty$ cioè $a_n \rightarrow 0$ opp. $a_n \rightarrow +\infty$

$$b_n \rightarrow \infty$$

$$\log a_n \rightarrow 0$$

$$\text{cioè } a_n \rightarrow 1$$

$$\text{f.i. } 1^\infty$$

Le f.i. allora sono: $+\infty - \infty$, $0 \cdot \infty$, $\frac{0}{0}$, $\frac{\infty}{\infty}$, 0^0 , $(+\infty)^0$, 1^∞

$$\log a_n b_n \quad b_n > 0 \quad \forall n, \quad a_n > 0 \quad \forall n, \quad a_n \neq 1 \quad \forall n,$$

$$\{a_n\} \text{ ref}, \{b_n\} \text{ ref}$$

$$\log a_n b_n = (\log a_n e) \left(\log b_n \right) \frac{\log b_n}{\log a_n}$$

$$1^\infty$$

$$a_n = \left(1 + \frac{1}{n}\right)^n \quad \text{è presente nelle f.i. } 1^\infty$$

Si può dim che $\{a_n\}$ è strett crescente e $a_n < 3 \quad \forall n$
quindi converge ad un numero < 3 che chiamiamo e

$$\text{Def} \quad a = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \quad \text{Si ha } e \in \mathbb{Q}$$

$$e > 1$$

Si può dimostrare che se $x_n \rightarrow \infty$ allora $\left(1 + \frac{1}{x_n}\right)^{x_n} \rightarrow e$

$$\text{es: } \left(1 + \frac{1}{n}\right)^n = \left[\left(1 + \frac{1}{\frac{n}{d}}\right)^{\frac{n}{d}} \right]^d \rightarrow e^d$$

se $d \in \mathbb{R}$

$$\left(1 + \frac{1}{n}\right)^n \rightarrow e^1$$

$$\left(1 - \frac{1}{n}\right)^n \rightarrow \frac{1}{e}$$

$$\left(\frac{n+4}{n+6}\right)^{2n+1} =$$

$$\left(\frac{2n+4}{n+6}\right)^{2n+1} \rightarrow +\infty$$

↓
que non si applica
nessuna regola

$$= \left[\left(1 + \frac{1}{x_n}\right)^{x_n} \right] = \left(\frac{n+4+2-2}{n+6}\right) = \left(\frac{n+6-2}{n+6}\right) = \left(1 + \frac{-2}{n+6}\right)^{2n+1} =$$

$$\left(1 + \frac{1}{\frac{n+6}{-2}}\right)^{2n+1} = \left[\left(1 + \frac{1}{\frac{n+6}{-2}}\right)^{\frac{n+6}{-2}} \right]^{\frac{-2(2n+1)}{n+6}} \rightarrow -4 \rightarrow e^{-4}$$

3

$$\left(\frac{n^2+2}{(n+1)^2}\right)^{n+3} = \left(\frac{n^2+2}{n^2+2n+1}\right)^{n+3} = \left(\frac{n^2+2n+1-2n-1+2}{n^2+2n+1}\right)^{n+3} =$$

$$= \left(1 + \frac{1-2n}{n^2+2n+1}\right)^{n+3} = \left[\left(1 + \frac{1}{\frac{n^2+2n+1}{1-2n}}\right)^{\frac{n^2+2n+1}{1-2n}} \right]^{\frac{(1-2n)(n+3)}{n^2+2n+1}} \rightarrow e^{-2}$$

Altrei limiti derivati dal numero e

$$1) \text{ se } e_n \rightarrow \frac{e^{e_n} - 1}{e_n} \rightarrow 1 \quad 2) \frac{\log(1+e_n)}{e_n} \rightarrow 1$$

$$3) \frac{(1+e_n)^d - 1}{e_n} \rightarrow d \quad (d \in \mathbb{R})$$

esempio:

$$n \cdot \log\left(\frac{n+1}{n}\right) = n \log\left(1 + \frac{1}{n}\right) = \frac{\log\left(1 + \frac{1}{n}\right)}{\frac{1}{n}} \rightarrow 1$$

Limiti notevoli con pure trigonometriche

$$\begin{aligned} & x \rightarrow 0 \quad \sin x \rightarrow 0 \quad \frac{\sin x}{x} \quad \text{f.i.} = \frac{0}{0} \\ & x \neq 0 \end{aligned}$$

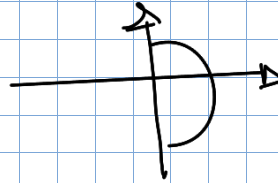
$$\text{Si ha } \forall x \in \left] -\frac{\pi}{2}; \frac{\pi}{2} \right[\quad |\sin x| \leq |x| \leq |\tan x|$$

$$\text{Se } x_n \rightarrow 0 \Rightarrow \left] -\frac{\pi}{2} < x_n < \frac{\pi}{2} \Rightarrow |\sin x_n| \leq |x_n| \leq |\tan x_n|$$

$$\Rightarrow 1 \leq \left| \frac{x_n}{\sin x_n} \right| \leq \frac{1}{|\cos x_n|}$$

$$\parallel$$

$$\frac{x_n}{\sin x_n} \parallel \frac{1}{\cos x_n}$$



$$\Rightarrow \cos x_n \leq \frac{\sin x_n}{x_n} \leq 1$$

$$\downarrow \quad \downarrow$$

$$1 \quad 1$$

$$\Rightarrow \frac{\sin x_n}{x_n} \rightarrow 1$$

$x_n \rightarrow 0$

$$\frac{\sin x_n}{x_n} \rightarrow 1 \quad \frac{x_n}{\sin x_n} \rightarrow 1 \quad \frac{\tan x_n}{\sin x_n} \quad \frac{1}{\cos x_n} \quad \frac{x_n}{\tan x_n} \rightarrow 1$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$1 \quad 1 \quad 1 \quad 1$$

$$\frac{\cos x_n}{\sin x_n} \rightarrow 1 \quad \frac{\sec x_n}{\tan x_n} \rightarrow 1$$

$$\frac{1 - \cos x_n}{x_n} = \frac{(1 - \cos x_n)(1 + \cos x_n)}{x_n(1 + \cos x_n)} = \frac{1 - \overset{x_n^2}{\cos^2 x_n}}{x_n(1 + \cos x_n)} = \frac{1}{1 + \cos x_n}$$

$$= \frac{\sin x_n}{x_n} \cdot \frac{\sin x_n}{1 + \cos x_n} \rightarrow 0$$

$$\frac{1 - \cos x_n}{x_n^2} = \frac{\sin^2 x_n}{x_n^2} \cdot \frac{1}{1 + \cos x_n} \rightarrow \frac{1}{2}$$

$$\downarrow \quad \downarrow$$

$$1 \quad \frac{1}{2}$$

Exemplo:

$$\frac{\lim_{n \rightarrow \infty} \frac{1}{2n+1}}{\frac{2}{3n}} = \frac{\lim_{n \rightarrow \infty} \frac{1}{2n+1}}{\frac{1}{\frac{3n}{2}}} \left(\frac{1}{2n+1} \cdot \frac{3n}{2} \right)$$

↓
L'Hôpital

$$\frac{\frac{1}{n+3}}{\lim_{n \rightarrow \infty} \frac{n+1}{n^2+4}} = \frac{\frac{n+1}{n^2+4}}{\lim_{n \rightarrow \infty} \frac{n+1}{n^2+4}} \left(\frac{n^2+4}{n+1} \cdot \frac{2}{n+3} \right) \rightarrow 1$$

↓
L'Hôpital

$$(2n^2+3) \lim_{n \rightarrow \infty} \frac{n}{n^3+1} = \lim_{n \rightarrow \infty} \frac{\frac{n}{n^3+1}}{\frac{n}{n^3+1}} \cdot \left(\frac{n}{n^3+1} (2n^2+3) \right) \rightarrow 2$$

↓
L'Hôpital

$$(3n-1) \lim_{n \rightarrow \infty} \frac{2n}{n^2+4} = \lim_{n \rightarrow \infty} \frac{\frac{2n}{n^2+4}}{\frac{2n}{n^2+4}} \left(\frac{2n}{n^2+4} (3n-1) \right) \rightarrow 6$$

↓
L'Hôpital

$$\frac{\lim_{n \rightarrow \infty}^2 \frac{2}{n+3}}{\lim_{n \rightarrow \infty} \frac{n}{3n^3+4}} = \frac{\lim_{n \rightarrow \infty}^2 \frac{2}{n+3}}{\left(\frac{2}{n+3} \right)^2} \cdot \frac{\frac{n}{3n^3+4}}{\lim_{n \rightarrow \infty} \frac{n}{3n^3+4}} \left(\left(\frac{2}{n+3} \right)^2 \frac{3n^3+4}{n} \right) \rightarrow 12$$

↓
L'Hôpital

$$\frac{1 - \cos \frac{2}{n+1}}{\lim_{n \rightarrow \infty} \frac{n+1}{n^3+2}} = \frac{1 - \cos \frac{2}{n+1}}{\frac{4}{(n+1)^2}} \cdot \frac{\frac{n+1}{n^3+2}}{\lim_{n \rightarrow \infty} \frac{n+1}{n^3+2}} \left(\frac{4}{(n+1)^2} \frac{n^3+2}{n+1} \right) \rightarrow \frac{1}{8}$$

↓
L'Hôpital